

Changes to gradient ascent with L2 for Stochastic Gradient Ascent algorithm

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1 The changes

Recall the final formula for derivative of the log likelihood function with L2

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i=0}^{n-1} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w)) - 2\lambda w_j$$

if $j > 0$

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i=0}^{n-1} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w))$$

if $j = 0$

For stochastic gradient ascent we will no longer sum over all data points. But we will do this for one data point. Or for a batch of data points of batch size B:

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i \in \mathcal{B}_t} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w)) - (\frac{2 \cdot B}{N})\lambda w_j$$

if $j > 0$

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i \in \mathcal{B}_t} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w))$$

if $j = 0$

The coefficient update will be:

$$w_j^{t+1} = w_j^t + \eta \left(\frac{\partial \ell(w^t)}{\partial w_j} \right) \left(\frac{1}{B} \right)$$

On beforehand data must be shuffled, to avoid that the algorithm is pushed in a biased direction by implicit ordering of data points.

2 Log likelihood, objective function and measure convergence

Recall the log likelihood function, which is essentially our objective function.

$$\ell(w) = \sum_{i=0}^{n-1} ((\mathbb{1}[y_i = +1] - 1)w^\top h(x_i) - \ln(1 + e^{-w^\top h(x_i)})) - \lambda \|w\|_2^2$$

For stochastic gradient ascent we will now compute:

$$\ell(w) = \sum_{i \in \mathcal{B}_t} ((\mathbb{1}[y_i = +1] - 1)w^\top h(x_i) - \ln(1 + e^{-w^\top h(x_i)})) - \left(\frac{B}{N}\right) \lambda \|w\|_2^2 \quad (\text{i})$$

And (i) will be averaged for the last k minibatches. This is used for assessing convergence.